

Renke (Richard) Han | PhD

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Profile

Deep knowledge of global xEV market trends (E-Drivetrain, zonal architectures, and EEA). More than 10 years of experience in power electronics hardware and software, with a track record of cross-functional collaboration in multicultural environments. Strong business orientation for design-wins and a holistic system perspective:

- E-Drivetrain application lead, reporting directly to the head of the product line with revenue above 1B
- Strong understanding of xEV OEM and Tier 1 technology trends and evolution, with extensive engagement in global customers for E-Drivetrain platforms and active relationships with leading OEMs and Tier 1
- Next-generation automotive microcontroller definition and portfolio management for E-Drivetrain applications
- Deliver fit-for-purpose solutions for stakeholders across different market segments, balancing cost and performance

Professional Experience

Staff -> Senior Staff -> Principal Engineer - Application lead and AURIX system architect <i>Automotive(ATV) · Infineon Technologies AG · Germany</i>	2023.06 – present
Senior -> Staff System Engineer <i>Green Industrial Power (GIP) · Infineon Technologies AG · Germany</i>	2021.08 – 2023.05
Postdoctoral Research Associate <i>Department of Engineering Science · University of Oxford · United Kingdom</i>	2018.11 – 2021.07
Departmental Tutor for Semiconductor Devices <i>Department of Engineering Science · University of Oxford · United Kingdom</i>	2020.10 – 2021.07
Visiting Scholar <i>Automatic Control Laboratory · École Polytechnique Fédérale de Lausanne · Switzerland</i>	2017.02 – 2017.08

Education

Ph.D. in Power Electronics Systems <i>Department of Energy Technology · Aalborg University · Denmark</i>	2015.11 – 2018.10
M.A.Sc. in Control Theory & Control Engineering <i>School of Information Science & Engineering · Northeastern University · China</i> GPA: 90/100, ranking: 1/134	2013.09 – 2015.07
B.Eng. in Automation <i>School of Information Science & Engineering · Northeastern University · China</i> GPA: 88.8/100, ranking: 14/270	2009.09 – 2013.07

Selected Experience

E-Drivetrain application lead and AURIX system architect <i>xEV application: Traction Inverter, On-board charger, HV/LV DCDC, X-in-1</i>	2023.06 – present
◦ Lead next-generation automotive MCU platform definition and customer requirements collection - Apply application expertise and a holistic view of market trends to balance customer requirements and die size (cost), defining technically fit products - Lead alignment across design experts, MCU architects, product marketing, and senior management to drive technical and business decisions - Peripheral performance definition: GTMv5 definition including special PWM signals(e.g., diode emulation, matrix converter modulation schemes), ADC concept(e.g., SAR, DSADC, Fast comparator) , interconnection (cross-bar), core architecture definition/selection for real-time application, functional safety concept and SW architecture ◦ Engage with customers across EMEA, Japan, and Greater China through on-site technical discussion, technology days, RFI responses, and requirements collection, driving significant design-wins and business-wins ◦ Develop successful go-to-market strategies for the AURIX family - Market trends and MCU selection guideline for OBC + DCDC (Read the guideline white paper) - Parallel processing unit enabling e-Drivetrain system BOM cost optimization and more X-in-1 options (Watch the webinar) - Analyze competitor product features (Tx, Rx, Nxx) and generate technical competitive guides ◦ Lead innovation projects as part-time project manager, driving a 20-person team from concept to delivery - Real-time junction-temperature estimation for traction inverters using the AURIX TC4xx parallel processing unit - New OBC + DCDC architecture by using one MCU, validation through hardware-in-loop setup (Read white paper) - Serve as firmware architect for a single-stage OBC reference design and a new traction-inverter architecture	

Industrial Reference Design and Digital Service

<i>Industrial application</i>	2021.08 – 2023.05
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- 22kW General Purpose Motor Drive firmware development ([View the reference design](#))
- IGBT power module real-time status monitor firmware development (Junction temperature, remaining lifetime)
- Flyback auxiliary power supply design (100 W, input DC link voltage 650V, output voltage: 15V and 24V)
- Design signal chains and PCBs for several applications, including T-sense modules and VCE-saturation measurement

Project: Robust Extra-Low-Cost Nano-Grids for the Developing World

Funded by UK Engineering and Physical Sciences Research Council (EPSRC)

2018.08 – 2021.07

- Developed scalable 12–200V, 300W multi-port isolated DC-DC power converters for PV-battery systems ([Watch the interview](#))
 - Selected topologies and passive components; designed high-frequency magnetic components
 - Developed embedded firmware for communication (UART/I2C/SPI), measurement (ADC and op-amps), digital control, Li-ion battery charge/discharge, cell balancing, and protection strategies on STM32
 - Built a Nano-Grid prototype with converters, ten 240 Wh Li-ion batteries, and 7.2 kWh of lead-acid batteries
 - Designed qualification tests covering measurement accuracy, safety, and reliability
- Coordinated with industrial (Tropical Power Cooperation) and academic (Cardiff University) partners
- Deployed 10 designed converters in a village in Africa, providing electricity for daily needs

Project: Pure software solution for power line communication (UK patent)

Funded by University of Oxford and Oxford Innovation Institute

2020.06 – 2021.03

- Achieved 2 kbps communication with an 8 kHz sampling frequency using amplitude modulation in the frequency band above the control bandwidth and below the sampling frequency
- Implemented software-only modulation and demodulation with Manchester encoding

Consultant Project: 5kW inductive Charger for Electrical Vehicle

Funded by University of Oxford

2020.09 – 2020.11

- Performed analysis and simulation of a weakly coupled transformer and its equivalent circuit
- Analyzed a tuned LCL circuit, including the LCL load-resonant converter and secondary pickup circuit
- Calculated the maximum transformer-winding current and power-transfer capacity to guide practical design

Awards and Prizes

Representative (1/7) of University of Oxford , Global Young Scientist Summit ·Singapore	2021.01
IEEE ECCE Travel Grants ·Cincinnati ·US	2017.10
Best Oral Presentation at IEEE 42nd IECON ·Florence ·Italy	2016.11
Outstanding Master Thesis at Provincial-level ·China	2015.06
Outstanding National Scholarship ·China	2014.06

Skills

Core competencies: systems architecture & product definition, cross-functional stakeholder management, go-to-market strategy, project/program management

Technical skills: Altium Designer, PLECS, C, AURIX, TI C2000, STM32F103, LTspice, and Eagle

Languages: Chinese (native), English (business proficiency), German (B1)

Other Experience

Member of the IEEE Power Electron., and Ind. Electron. Societies	2015.11–
President of Chinese Students and Scholars Association, Denmark	2016.02 - 2017.08

Selected Top Publications

1 UK patent, 15 journal papers (5 as first author in IEEE Transactions on Power Electronics, Power Systems, and Industry Applications), and 11 conference papers (oral presentations)

Google Scholar Citations: 2249, H-index: 16 i10-index: 20

1. **R. Han**, M. Moniruzzaman, and M. N. Münzer, “Embedded implementation of a thermal coupling matrix using aurix tc4x parallel processing unit for xev applications,” in *PCIM Europe 2026; International Exhibition and Conference for Power Electronics, Intelligent Motion, Renewable Energy and Energy Management*, 2026, pp. 1–8.
2. **R. Han** and D. J. Rogers, “Zero-additional-hardware power line communication for dc–dc converters,” *IEEE Transactions on Power Electronics*, vol. 37, no. 11, pp. 13 107–13 118, 2022.
3. **R. Han**, L. Meng, J. M. Guerrero, and etc, “Distributed nonlinear control with event-triggered communication to achieve current-sharing and voltage regulation in dc microgrids,” *IEEE Transactions on Power Electronics*, vol. 33, no. 7, pp. 6416–6433, 2018.
4. E. Shelton, **R. Han**, D. Rogers, J. Carter, L. Louco, M. Beadman, and P. Palmer, “Comparison of fast switching high current power devices,” in *PCIM Europe digital days 2021; International Exhibition and Conference for Power Electronics, Intelligent Motion, Renewable Energy and Energy Management*, 2021, pp. 1–8.